ALMA MATER STUDIORUM
UNIVERSITÀ DI BOLOGNA

Dispatching mechanisms and algorithms

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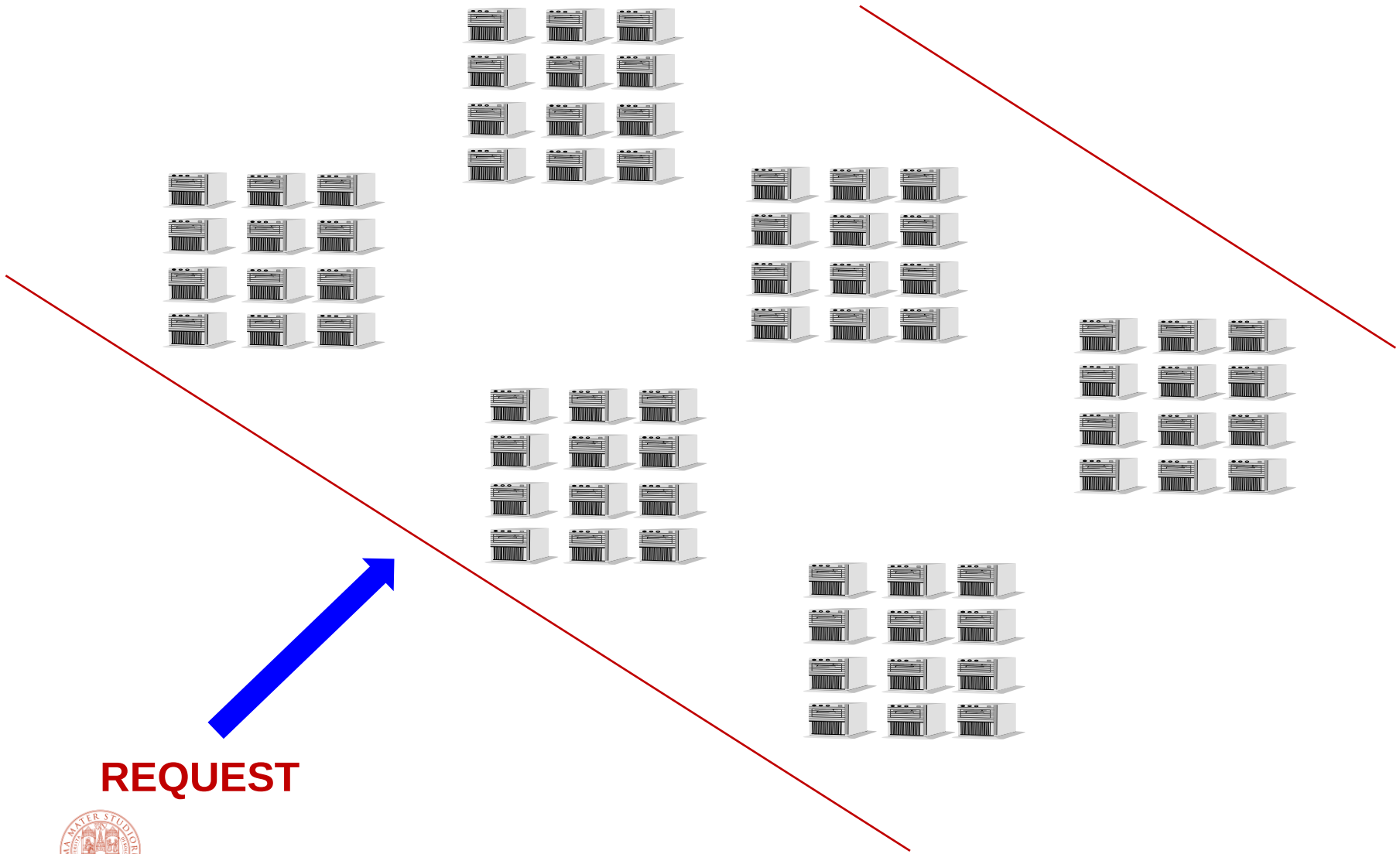
The dispatching problem

Resource selection

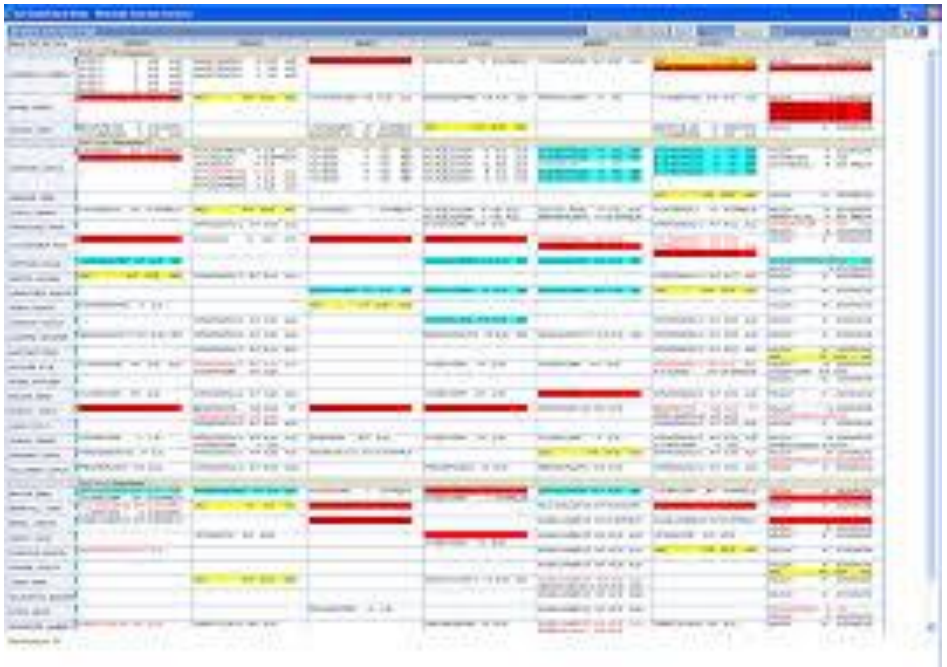


The theory comes from scheduling

- **Offline scheduling**
- **Online scheduling**
- Scheduling for **hard real-time systems**

Offline scheduling

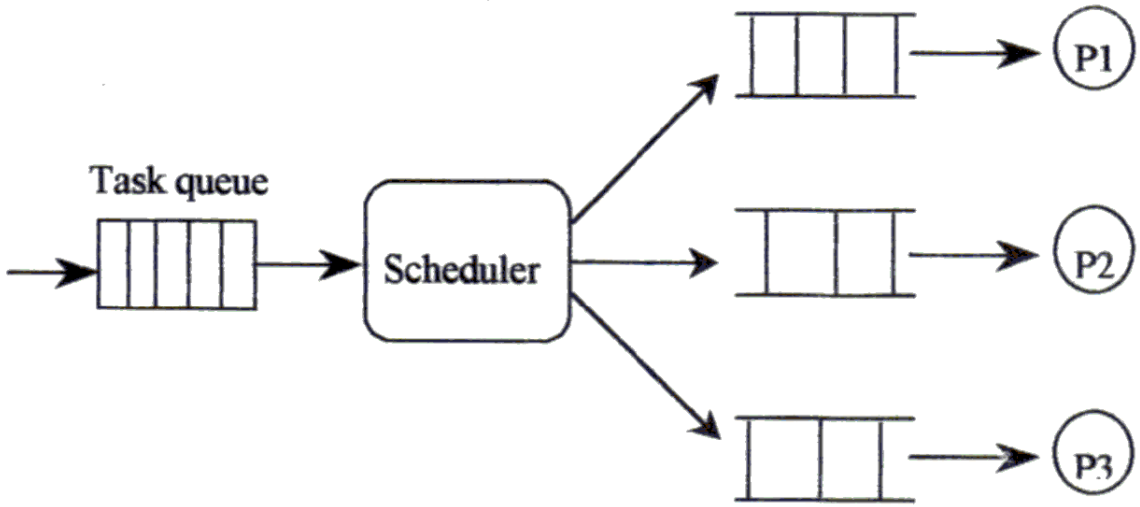
- In **offline scheduling** problems, there is no conception of “point of time”. Algorithms stay outside the real-time world and can see the past and future of tasks
- Hence, the algorithms know the total knowledge of the problem before we make decision → ***optimization is a possible goal***
- Examples: timetable, crew assignment, guard duty, industrial production, Gantt chart, etc.



Online scheduling

- **Online scheduling** is characterized by making decision “online”, which means a point in the time axis. In this point of time, the algorithms cannot see the future jobs or tasks, whereas they only know the jobs or tasks before or at this point of time → ***just approximation of optimum is possible***

- **Temporal term of scheduling:** Long-term, Mid-term, Short-term



Online scheduling

An **online scheduling algorithm** decides which requests are given access to resources from moment to moment, e.g.,

- **Operating systems**: multiple processes competing for the CPU
- **Hypervisor**: virtual machines
- **Router**: packets competing for network links
- **RAID drive**: read/write operations competing for disks
- **Print manager**: prints

Possible goals of online scheduling

- Minimizing (average) response time
- Maximizing throughput
- Maximizing availability
- Fairness in sharing resources
- Meeting deadlines
- Avoiding starvation
- Load balancing
- Load sharing
- …
- Combination of two or multiple goals

Online scheduling algorithms

1. ***First Come First Served*** (FCFS)
2. ***Round Robin*** (RR)
3. ***Round Robin with quantum***
4. ***Weighted Round Robin*** *(WRR)*
5. ***Shortest Job First*** (SJF), also known as ***Shortest Time to Completion First*** (STCF)
6. ***Shortest Remaining Time First*** (SRTF), also known as ***Shortest Remaining Time to Completion First*** (SRTCF), that is the preemptive version of SJF
7. ***Priority*** (with and without preemption)
8. ***Multi-level***: multiple queues, each with its scheduling algorithm
9. ***Multi-level with fairness***

Scheduling and Dispatching

- **Dispatching** is more related to *short-term online scheduling* algorithms, hence in our contexts we prefer to use *dispatching* instead of *scheduling*
- There are many definitions and interpretations that are applicable to different contexts. My favorite:
  - **Dispatching is "what next and where"**
  - **Scheduling is "what, where, when, why"**

Classification

- **Dispatching contexts**
  - Local area
  - Geographical area
- **Dispatching mechanisms - *local***
  - Switch (local)
    - Layer 4
    - Layer 7
- **Dispatching algorithms - *geo***
  - DNS resolution (geo)
  - HTTP redirection (geo)

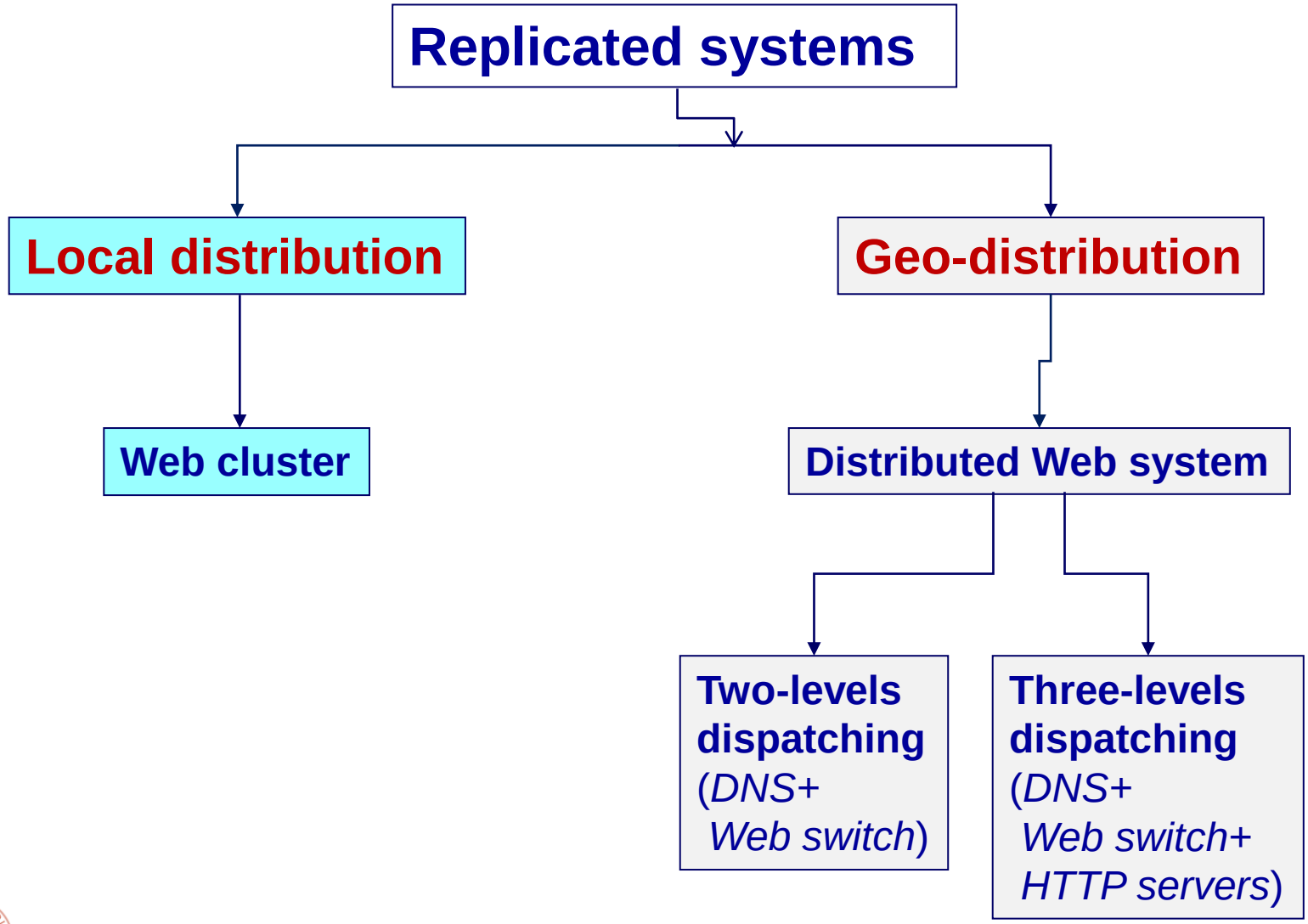
Dispatching algorithms

- Static
- Dynamic
- Adaptive

Dispatching algorithms

- Static
- Dynamic
- Adaptive

Taxonomy


Replicated systems
Local distribution
Geo-distribution
Web cluster
Distributed Web system
Two-levels dispatching (DNS+ Web switch)
Three-levels dispatching (DNS+ Web switch+ HTTP servers)

Dispatching mechanisms

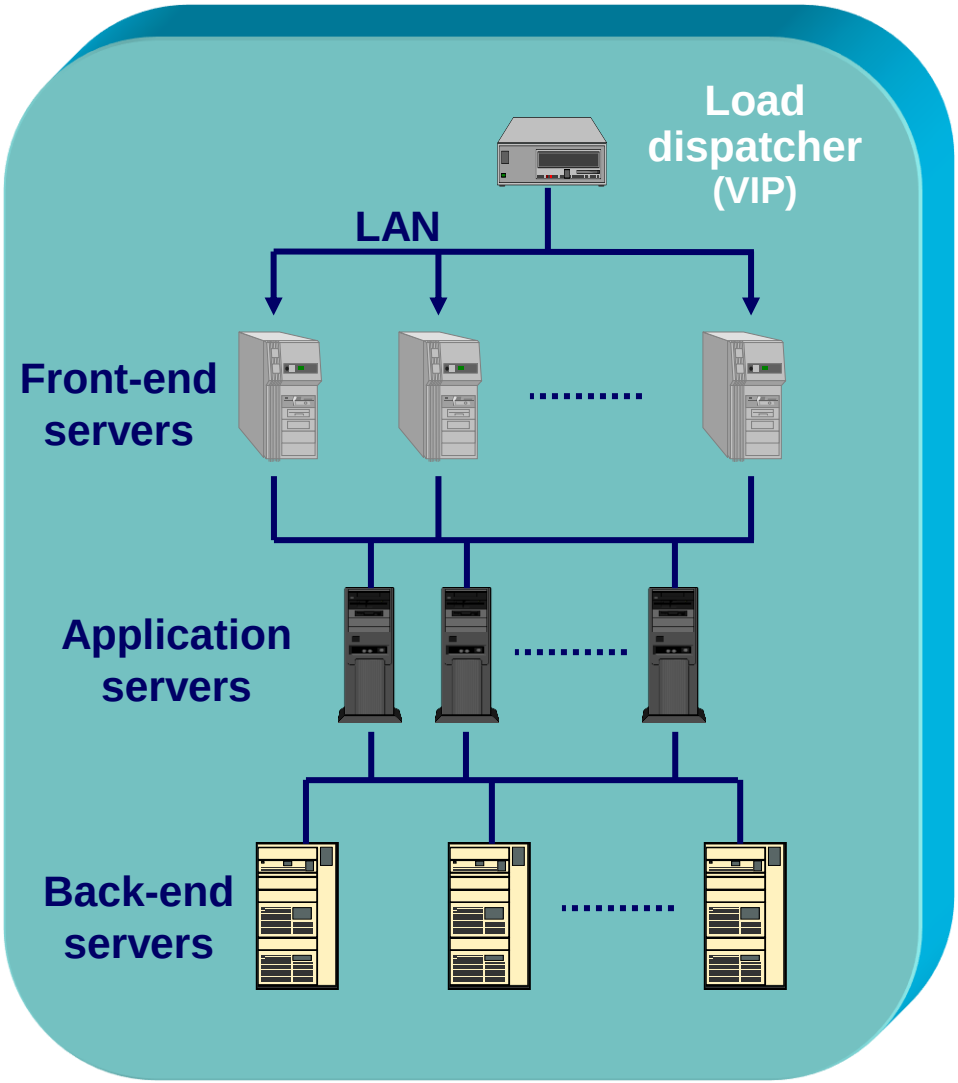
	Context
1. **Switch dispatching**	***Local***
2. **DNS dispatching**	***Global***
3. **HTTP dispatching** (*HTTP redirection*)	***Global***

Local dispatching

Locally distributed system

A locally distributed architecture has

- One hostname (e.g., “www.unibo.it”)
- One IP address (*virtual IP address*) of the front-end switch
  - Only its IP address is visible and corresponds to the site IP address)
  - All internal servers have hidden IP addresses



1. Switch dispatching

- The dispatcher of a Web cluster has total control on all requests reaching the cluster

- It can use **client** information, **server** information or **client and server** information
  - Client information can be ***content-blind*** (switch at layer 4)
  - Client information can be ***content-aware*** (switch at layer 7)

Server info-aware algorithms

- Request assignment based on *server load info*
  - **Least loaded server** (← AVOID!)
  - **Random(*k*)+least loaded**: pick *k* servers randomly and choose the least loaded
  - **Weighted Round-Robin** (**WRR**)
    - it allows configuration of weights as a function of the server load
- Possible metrics to estimate the server load
  - **Input metrics**: information get by the Web switch without server cooperation, e.g., *Active connections*
  - **Server metrics**: information get by the servers and transmitted to the Web switch, e.g., *CPU/Disk utilization, response time*
  - **Forward metrics**: information get directly by the Web switch, e.g., *emulation of requests to servers*

Client info-aware algorithms

Session identifiers

- HTTP requests with same ***SSL id*** or same ***cookie*** assigned to the same server
  - Goal: avoid multiple client identifications for the same session

Content partition

- Content partitioned among servers according to **file type** (HTML, image, dynamic content, audio, video, …)
  - Goal: use specialized servers for different contents
- Content partitioned among servers according to **file size** (Thresholds may be chosen dynamically)
  - Goal: augment load balancing
- File space partitioned among the servers through a hash function
  - Goal: improve *cache hit rate* in Web servers

Client info-aware algorithms

Content Aware Partition (CAP)

- Resource classification according to the impact of HTTP requests on main server components, e.g.,
  - *Low impact* (small-medium static files)
  - *Network bound* (large file download)
  - *Disk bound* (database queries)
  - *CPU bound* (“secure” requests)
- Cyclic assignment of each class of requests to servers
- Goal: to augment load sharing of *component bound* requests among servers

Dispatching algorithms

1. **Static** → *information-less*

2. **Dynamic** → *they get continuous information from the environment*
   - **Client info-aware**
   - **Server info-aware**
   - **Client info-aware and Server info-aware**

3. **Adaptive** → *they use different algorithms and apply the "best" one depending on system's status*

Static algorithms

- **Random**
  - no information about the system state
  - no history about previous assignments

- **Round Robin (RR)**
  - no information about the system state
  - history is about the previous assignment only

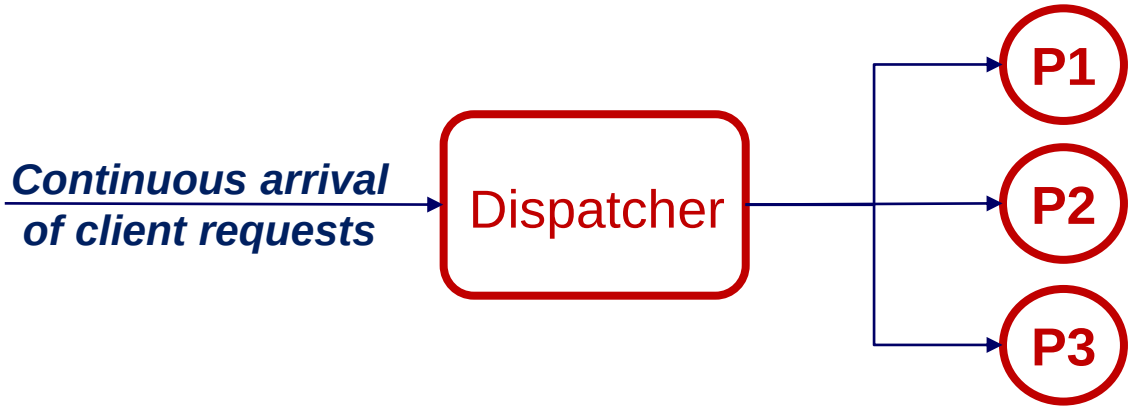
Dynamic: possible *client* information

1. IP address
2. Source port
3. Novel request (TCP syn)
4. HTTP *vs* HTTPS connection
5. Type of request (URL)
6. Cookie

Client info-aware algorithms

- **Client partition**
  - Request assignment based on client information in inbound packets
    - Client IP address
    - Client port

Dynamic: possible *server* information



1. Information about servers known at the dispatcher level
   - Number of active requests
   - Number and type of active requests
2. Information provided by the servers
   - Length of process queue
   - Resource utilization (CPU, disk, network, memory)
3. Information measured by the dispatcher
   - Response time (measured through request probes)

Server info-aware algorithms

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Other combinations

- **Client info-aware + Server info-aware**

- **Adaptive**: dynamic algorithms that update their decisions based on significant changes in system's conditions

Client info-aware algorithms

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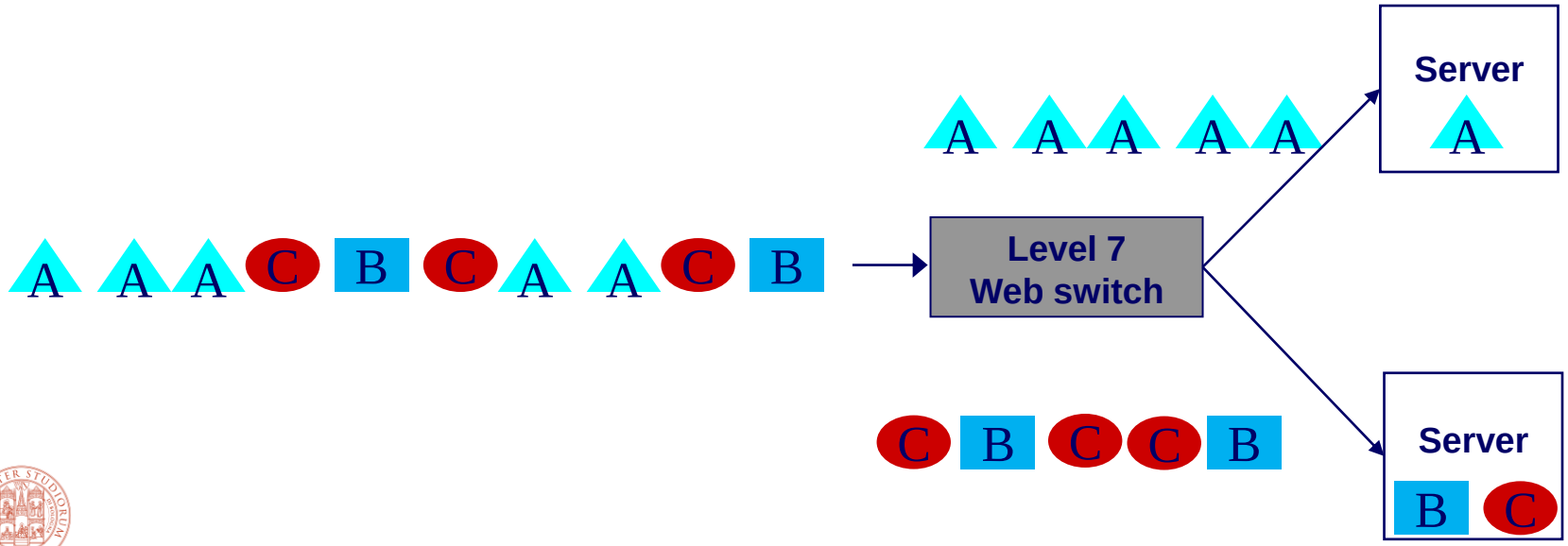
  - Cyclic assignment of each class of requests to servers
  - Goal: to augment load sharing of *component bound* requests among servers

Client and server info-aware algorithm

Locality-Aware Request Distribution (LARD)

- First request for a given target assigned to the least loaded server (metrics: *number of active connections*)
- Subsequent requests for the same target assigned to the previously selected server
- Goal: improve locality (*cache hit rate*) in server cache



Geo dispatching

Geographically distributed system

- ***Web-based service realized on an architecture of geographically distributed Web clusters***
- **Web site addresses**
  - One hostname (e.g., “www.site.com”)
  - One IP address for each Web cluster

First level dispatching

The authoritative DNS, at the lookup, selects one Web cluster

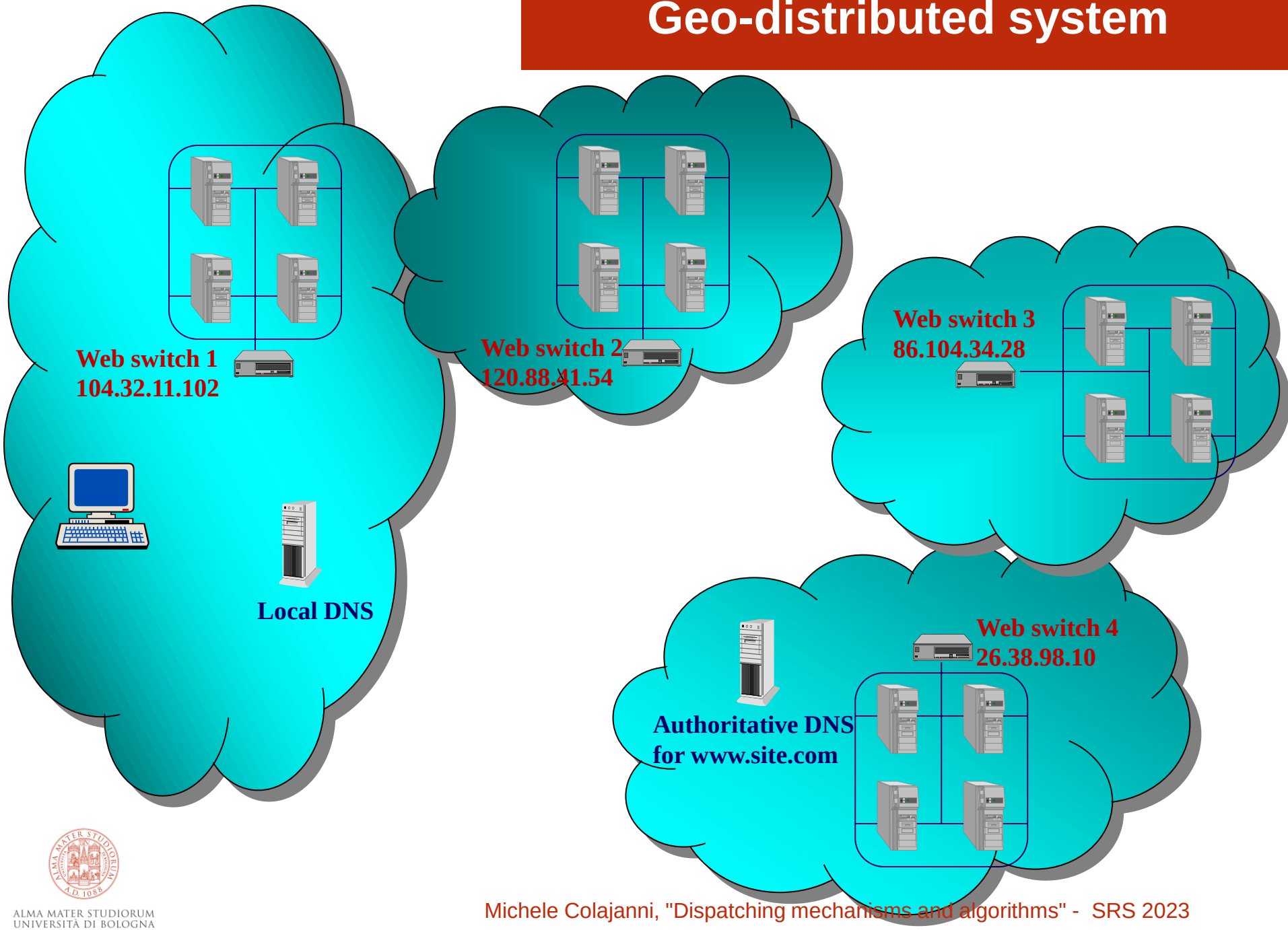
Second level dispatching

The Switch of the Web cluster selects one internal server

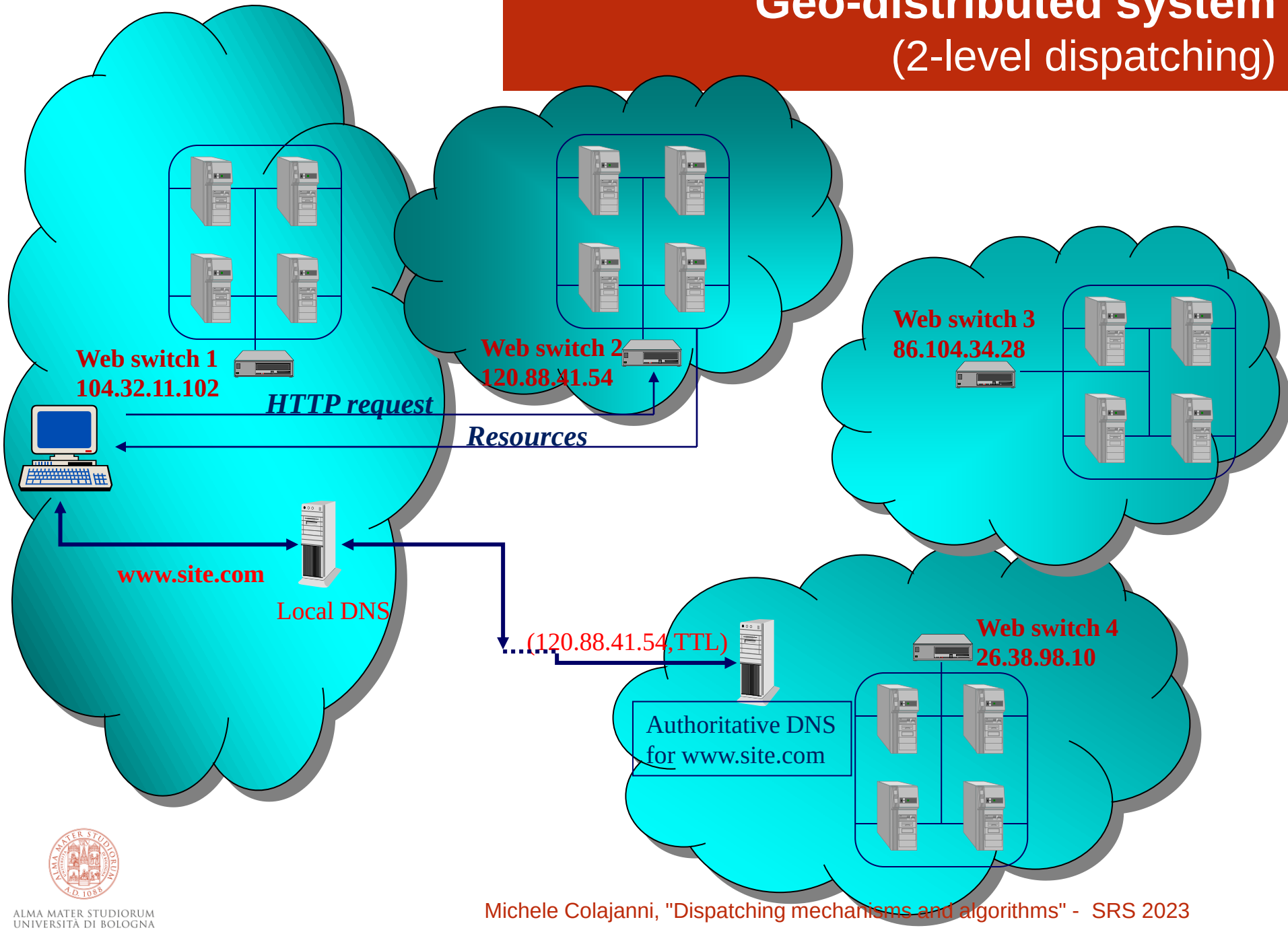
Third level dispatching

The Switch (layer 7) may redirect the received request to another Web cluster

Geo-distributed system



Geo-distributed system
(2-level dispatching)



DNS dispatching

- The distributed Web cluster architectures implement global dispatching by intervening in the ***lookup phase*** of the address request:
  - A client asks for the **IP address** of a Web server corresponding to the **hostname** in the URL
  - If the hostname is valid, it receives the couple

    (IP address, TimeToLive)

- The ***enhanced authoritative DNS*** of the Web site (or *another entity* that replaces or integrates the authoritative DNS) can use various dispatching policies to select the “best” Web cluster

Issues of DNS dispatching

Typical issues

- Load spikes in some hours/days

Additional issues

- Traffic depending on time zones
- Client distribution among Internet zones (AS)
- Proximity between the client and the Web cluster
- Caching of [hostname-IP] at intermediate DNS servers for TTL interval

All modern geographically distributed architectures use some sort of DNS dispatching

Issues of DNS dispatching

1. **[Hostname - IP address] caching problem**: the authoritative DNS of highly popular Web-based services controls only 5-7% of traffic reaching the servers of the site

2. **Internet proximity problem**: which is the cluster closest to the client? Is the cluster closer to the client always the best choice?

Unlike the switch (controlling 100% traffic), the DNS should use sophisticated dispatching algorithms

Caching issue: actions on TTL values

- **Constant TTL**
  - Set TTL=0 to augment the DNS control
  - Drawbacks
    - Non-cooperative name servers (they may ignore small TTL values <300 seconds)
    - Browser caches
    - Risk of overloading authoritative name servers

- **Adaptive TTL**
  - Tailor TTL value adaptively for each address request by taking into account the popularity of client Internet domain and Web server loads

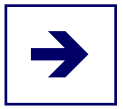
Internet proximity issue

- **Internet proximity is an interesting issue**

  Client-server *geographic proximity* does not mean *Internet proximity* (*round trip latency*)

  - ***Static* information**
    - client IP address to determine Internet zone (geographical distance)
    - hop count ("stable" more than "static" information)
      - *network* hops (e.g., `traceroute`)
      - *Autonomous System* hops (routing table queries)

It does not guarantee selection of the best connected Cluster because "links are not created equal"



Internet proximity

- ***Dynamic* evaluation of proximity**
  - round trip time (e.g., `ping`, `tcping`)
  - available link bandwidth (e.g., `cprobe`)
  - latency time of HTTP requests (request emulation)

Additional time and traffic costs for evaluation

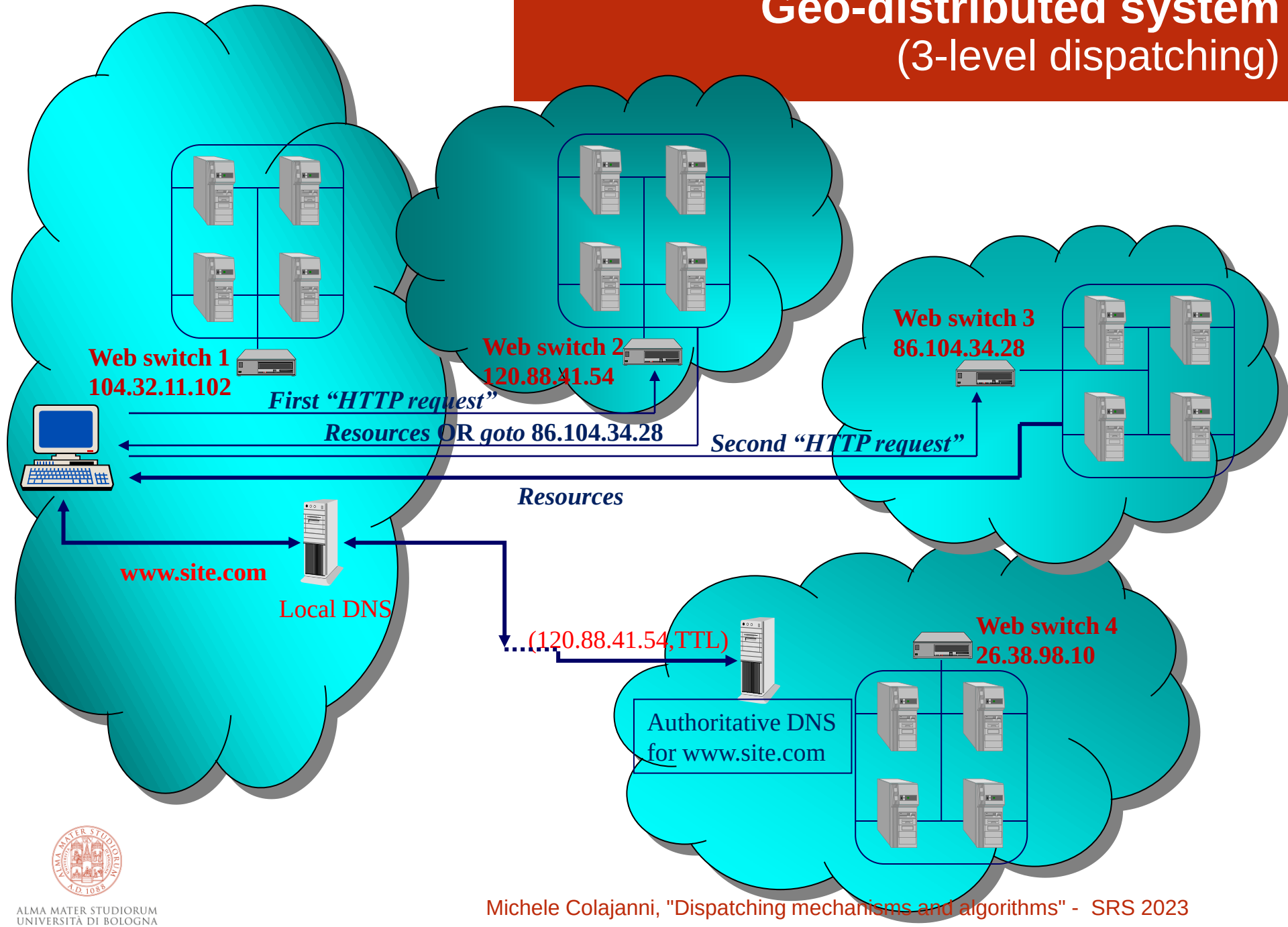
A related open issue:

Is there correlation between hop counts and round-trip time?

- "Old" measures: close to zero ['90]
- "Recent" measures: strong, reasonably strong

WHY?

Geo-distributed system
(3-level dispatching)



HTTP redirection mechanism

- The redirection mechanism is part of the HTTP protocol and is supported by current browser and server software
- DNS and Web switch use centralized scheduling disciplines
- HTTP redirection is a distributed scheduling policy, in which all Web clusters independently can participate in re-assigning requests
- Redirection is transparent to the user (**not to the client**!)

HTTP redirection

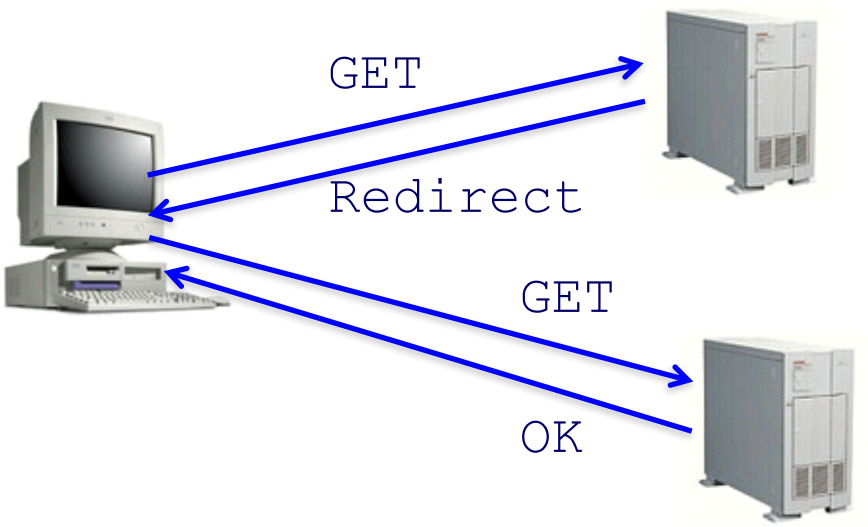
message header
HTTP OK status code
302 - "Moved temporarily" to a new location

- "New location"
  - Redirection to an IP address (**better performance**)
  - Redirection to an hostname
- *HTTP redirection* is easy to implement but works for HTTP services only
- **As an alternative, *IP tunneling* can be used when it is necessary to provide other services**

Dispatching mechanisms: *summary*

Mechanism	*Executor*	*Item*
Hostname resolution	**DNS / Other entity**	**Session**
- *Local dispatching*		
- Global dispatching		
HTTP redirection	**Web server**	**Page request**
- *Local dispatching*		
- Global dispatching		
Packet redirection	**Switch**	**Hit / Page request**
- Local dispatching		

Server selection mechanism: *HTTP redirection*



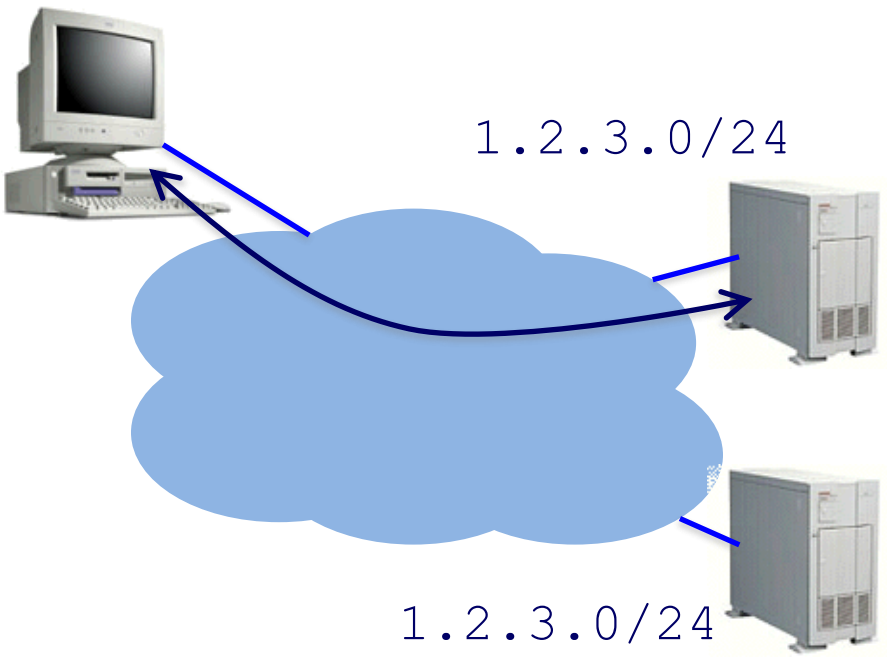
Advantages

- Fine-grain control
- Selection based on client IP address

Disadvantages

- Extra round-trips for TCP connection to server
- Overhead on the server

Server selection mechanism: *Anycast routing*



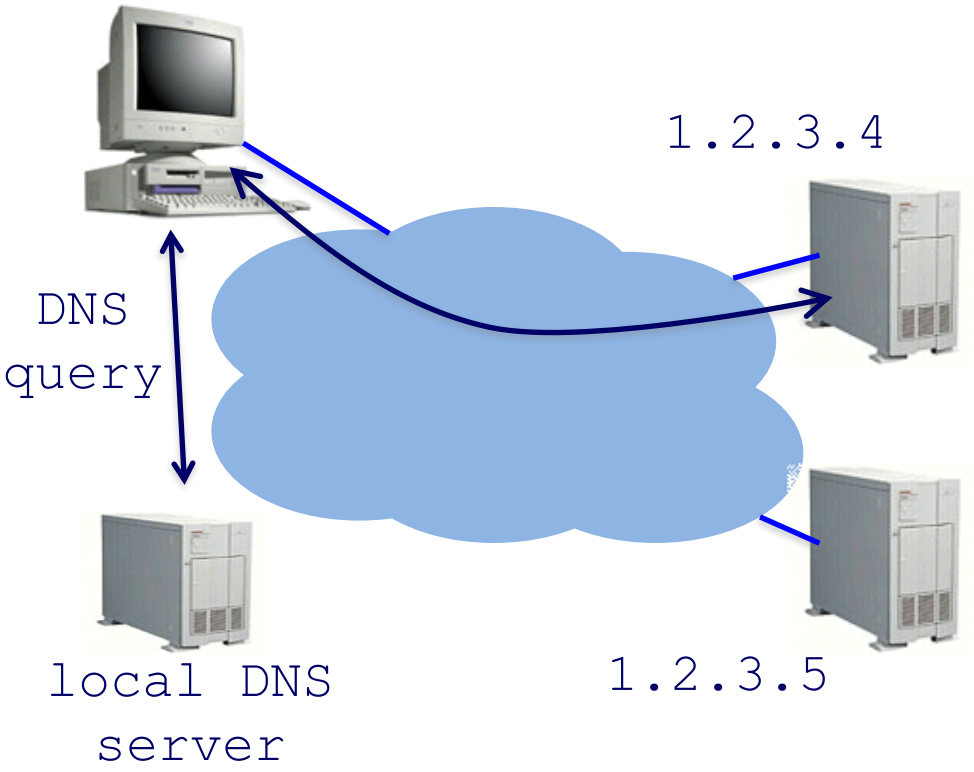
Advantages

- No extra round trips
- Route to nearby server

Disadvantages

- Does not consider server load
- Different packets may go to different servers
- Used only for simple request-response apps

Server selection mechanism: *DNS-based*



Advantages

- Avoid TCP set-up delay
- DNS caching reduces overhead
- Relatively fine control

Disadvantage

- Based on IP address of local DNS server
- DNS TTL limits adaptation with *hidden load* risks

Geographically distributed systems

- **Two-level dispatching**
  - Simplicity
  - High control on load reaching the Web cluster
  - Slow reaction to an overloaded Web cluster
- **Three-level dispatching**
  - Complex dispatching
  - Redirection valid only for HTTP-based services
  - Guarantees immediate actions to shift the load away from an overloaded Web cluster

This is a typical example where ***adaptive dispatching*** can work well → Use two-level dispatching when the system state is in good conditions; switch to three-level otherwise

Several alternatives

- Two-levels *vs.* Three-levels dispatching
- Which ***selection*** policy? *Redirect-All vs. Redirect-Heavy*
- Which ***location*** policy?

- Dispatching algorithms
  - Level 1 (*DNS*): e.g., proximity
  - Level 2 (*Switch*): e.g., Weighted Round Robin
  - Level 3 (*Web servers*)
    - Selection policy (“*which requests have to be redirected?”*): e.g.,
      - Redirect all requests
      - Redirect just “heavy” requests in terms of size or processing
    - Location policy (*“towards which system?”*): e.g.,
      - Round Robin, Least loaded system, System proximity, …
      - Combinations: pick

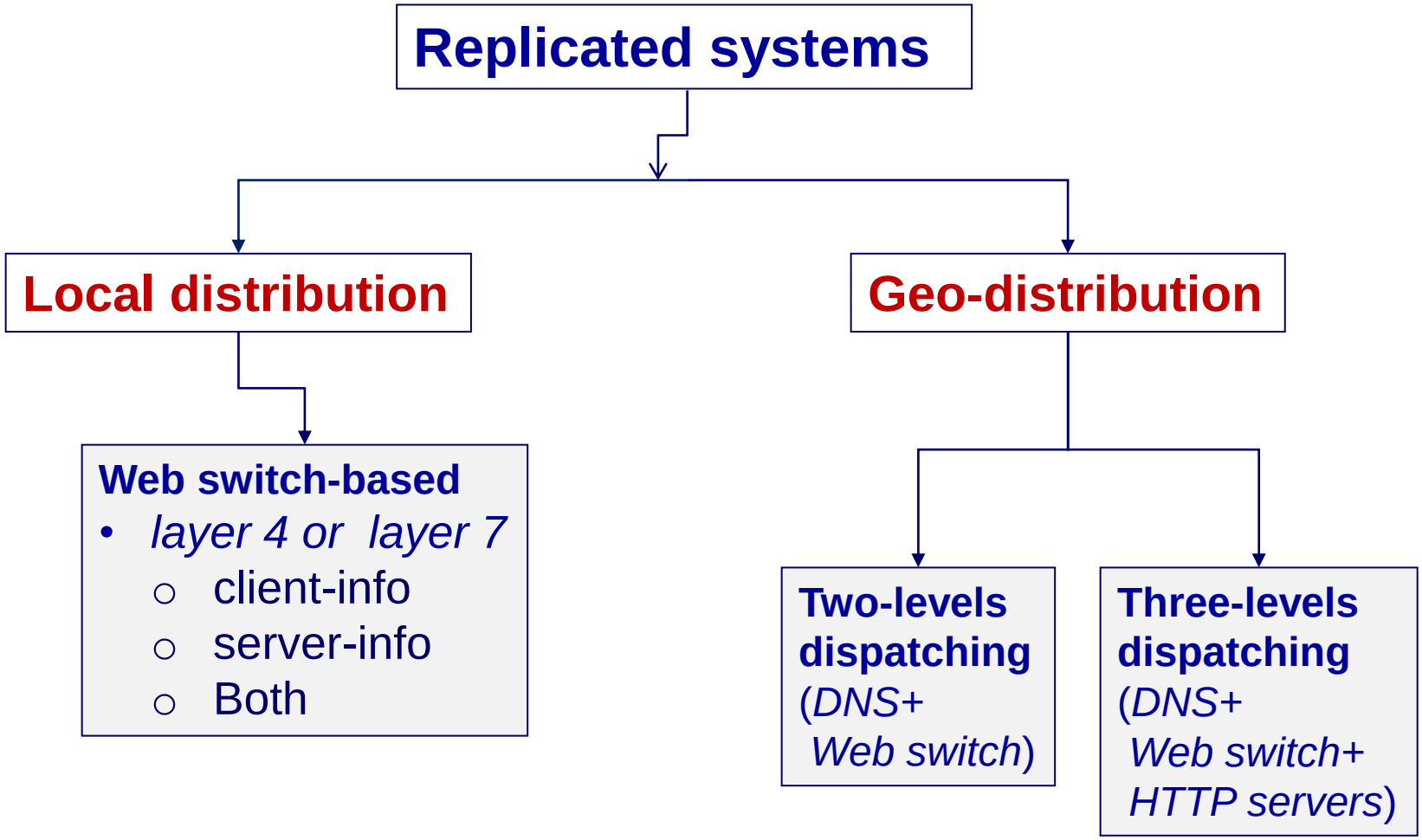
Server selection policies

- **Live server**
  - For availability

Requires continuous monitoring of liveness, load, and performance

- **Low load (not always the lowest)**
  - To balance load across the servers
- **Closest**
  - Nearest geographically, latency, or measured in round-trip time
- **Best performance**
  - Throughput, …
- **Others**
  - Cheapest bandwidth, electricity savings, …

Summary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